(Question 3b.)

$$A = \begin{pmatrix} 4 & -2 & 1 \\ 3 & 6 & -4 \\ 2 & 1 & 8 \end{pmatrix}$$

$$(A|I) = \left(\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 3 & 6 & -4 & 0 & 1 & 0 \\ 2 & 1 & 8 & 0 & 0 & 1 \end{array}\right)$$

$-\frac{3}{4}R1 + R2 \to R2$

$$\left(\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 0 & 7.5 & -4.75 & -0.75 & 1 & 0 \\ 2 & 1 & 8 & 0 & 0 & 1 \end{array}\right)$$

$-\frac{1}{2}R1 + R3 \to R3$

$$\left(\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 0 & 7.5 & -4.75 & -0.75 & 1 & 0 \\ 0 & 2 & 7.5 & -0.5 & 0 & 1 \end{array}\right)$$

$-\frac{2.0}{7.5}R2 + R3 \to R3$

$$\left(\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 0 & 7.5 & -4.75 & -0.75 & 1 & 0 \\ 0 & 0 & 8.77 & -0.3 & -0.2667 & 1 \end{array}\right)$$

$\frac{4.75}{8.77}R3 + R2 \to R2$

$\frac{x}{4} = \frac{17}{263}$

$$\left(\begin{array}{ccc|ccc} 4 & -2 & 1 & 1 & 0 & 0 \\ 0 & 7.5 & 0 & -0.912 & 0.854 & 0.542 \\ 0 & 0 & 8.77 & -0.3 & -0.2687 & 1 \end{array}\right)$$

$\frac{-1}{8.77}$ R1 + R1 → R1

$$\left(\begin{array}{ccc|ccc} 4 & -2 & 0 & 1.034 & 0.0341 & -0.114 \\ 0 & 7.5 & 0 & -0.912 & 0.854 & 0.542 \\ 0 & 0 & 8.77 & -0.3 & -0.27 & 1 \end{array}\right)$$

$\frac{2}{7.5}$ R2 + R1 → R1

1/8.77 R3 → R3

7.5 → R2

$$\left(\begin{array}{ccc|ccc} 4 & 0 & 0 & 0.791 & 0.258 & 0.0305 \\ 0 & 7.5 & 0 & -0.912 & 0.854 & 0.542 \\ 0 & 0 & 8.77 & -0.3 & -0.27 & 1 \end{array}\right)$$

1/4 → R1

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.1977 & 0.0645 & 0.00763 \\ 0 & 1 & 0 & -0.122 & 0.114 & 0.0123 \\ 0 & 0 & 1 & -0.034 & -0.031 & 0.114 \end{array}\right) \cong (I \mid A^{-1})$$

$$A^{-1} \approx \frac{1}{263}\begin{pmatrix} 51.995 & 16.9635 & 2.00669 \\ -32.086 & 29.982 & 19.0149 \\ -8.942 & -8.153 & 29.982 \end{pmatrix}$$